Solution:

Let the number be x.

The student had to find $\frac{8x}{9}$ but, he had found $\frac{x}{\left(\frac{8}{9}\right)}$, that is $\frac{9x}{8}$.

Now, $\frac{9x}{8}-\frac{8x}{9} = 34$ (given)

$$\frac{81x-64x}{72} = 34 \Rightarrow \frac{17x}{72} = 34$$

$$x = \frac{34\times 72}{17} = 144$$

Example 1.21

Simplify: $\left(\frac{4}{3}-\left(\frac{-3}{2}\right)\right)+\left(\frac{-5}{3}\div\frac{30}{12}\right)+\left(\frac{-12}{9}\times\frac{-27}{16}\right)$

Solution:

Here, $\left(\frac{4}{3}-\left(\frac{-3}{2}\right)\right)+\left(\frac{-5}{3}\div\frac{30}{12}\right)+\left(\frac{-12}{9}\times\frac{-27}{16}\right) = \left(\frac{4}{3}+\frac{3}{2}\right)+\left(\frac{-5}{3}\times\frac{12}{30}\right)+\left(\frac{-12}{9}\times\frac{-27}{16}\right)$

$$= \left(\frac{8}{6}+\frac{9}{6}\right)+\left(\frac{-1}{1}\times\frac{4}{6}\right)+\left(\frac{-3}{1}\times\frac{-3}{4}\right)$$

$$= \left(\frac{17}{6}\right)+\left(\frac{-4}{6}\right)+\left(\frac{9}{4}\right)$$

$$= \left(\frac{17-4}{6}\right)+\frac{9}{4} = \frac{13}{6}+\frac{9}{4}$$

$$= \frac{26+27}{12} = \frac{53}{12}$$

Exercise 1.2

1. **Fill in the blanks:**
   (i) The value of $\frac{-5}{12}+\frac{7}{15}=$ _____.
   (ii) The value of $\left(\frac{-3}{6}\right)\times\left(\frac{18}{-9}\right)$ is _____.
   (iii) The value of $\left(\frac{-15}{23}\right)\div\left(\frac{30}{-46}\right)$ is _____.
   (iv) The rational number _____ does not have a reciprocal.
   (v) The multiplicative inverse of –1 is _____.
2. **Say True or False:**
   (i) All rational numbers have an additive inverse.
   (ii) The rational numbers that are equal to their additive inverses are 0 and –1.
   (iii) The additive inverse of $\frac{-11}{-17}$ is $\frac{11}{17}$.

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